

AKASH GOJURU

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Undergraduate junior with a strong interest in the hardware industry, particularly in digital design and verification. I am currently seeking an internship opportunity to gain hands on experience in this field. I am actively building my skills in Computer Architecture and organization, as well as hardware description languages including Verilog and VHDL.

Education

Iowa State University (ISU) Aug 2024 – (Expected) Dec 2026
Bachelor of Science in Computer Engineering (BS-CPRE) GPA: 3.60/4.0

Coursework: Computer Architecture and Organization, Digital Logic Design, Electronic Circuits and Systems, Embedded System Design, Advanced Programming Languages and Introduction to the Design and Analysis of Algorithms.

Amity University (AU) Aug 2022 – May 2024
Bachelor of Engineering in Computer Science and Engineering (BE-CSE) CGPA: 8.64/10

Coursework: Calculus I & II, Introduction to Data Structures, Statistics, Materials and Semiconductors.

Technical Skills

Programming Languages: VHDL, Verilog, C++, Java, C, Python,

Fundamental Knowledge: RISC-V, MIPS, ASIC Design Flow, CPU, Linux, Finite State Machine, RISC-V Assembly, Git, Bash, x86, SPI/UART/I2C

Software Tools: QuestaSim, ModelSim, Code Composer Studio, (CCS), Quartus Prime, Xilinx Vivado

Projects

RISC-V 5-Stage Pipelined Processor (VHDL) Oct 2025 – Nov 2025

- Implemented a 5-stage pipelined RISC-V CPU (IF, ID, EX, MEM, WB) with forwarding logic, hazard detection, and stall control.
- Added branch handling, bypass networks, and pipeline registers to achieve cycle-accurate execution.
- Tested the processor using RARS-generated assembly (.s) files and automated verification scripts to ensure pipeline correctness and hazard resolution.
- Used ModelSim/QuestaSim to analyze waveforms, verify pipeline timing, and debug structural and control hazards.
- Integrated ALU, register file, control unit, and memory modules into a modular RTL design, enabling scalable extension for future ISA features and performance optimizations.

RISC-V Single-Cycle Processor (VHDL) Sep 2025 – Oct 2025

- Designed a full single-cycle RISC-V processor supporting arithmetic, logical, branch, and memory instructions.
- Built core components including ALU, register file, control unit, immediate generator, PC logic, and memory interface.
- Validated the design on FPGA hardware using pushbuttons, switches, and LEDs for I/O.
- Verified full instruction functionality by running RARS .s test programs through automated test scripts, validating register updates and execution correctness.
- Debugged timing, control signal interactions, and instruction sequencing using ModelSim/QuestaSim waveform analysis.

Pipelined Multiply-Accumulate (MAC) Unit (VHDL) Aug 2025 – Sep 2025

- Designed a pipelined MAC unit with weight registers for TPU-like systolic array architectures.
- Implemented the architecture in structural VHDL using modular multiplier, adder, and pipeline components.
- Verified functionality using QuestaSim testbenches and waveform analysis.
- Optimized and debugged the RTL to achieve correct cycle-accurate behavior and improved throughput.

Work Experience

Tech Support Assistant, Iowa State University (CELT) Jan 2025 – Present

- Managed classroom recording systems and resolved real-time technical issues.
- Edited and processed instructional videos to enhance clarity and accessibility.
- Provided software and technical support to faculty and staff across multiple platforms.

Leadership & Community Involvement

Teaching Assistant, Amity University Jan 2023 – May 2023

Applied Mathematics – I (Linear Algebra & Multivariable Calculus)

- Assisted with instruction, grading, and classroom support for over 50 students in applied mathematics.
- Held weekly office hours to provide individual tutoring in linear algebra and multivariable calculus.

Chip Forge Club, Iowa State University Aug 2025 – Present

Member

- Contributed to ASIC design projects involving RTL development, verification, fabrication, and bring-up.
- Participated in digital and analog circuit design while learning standard EDA tools and workflows.